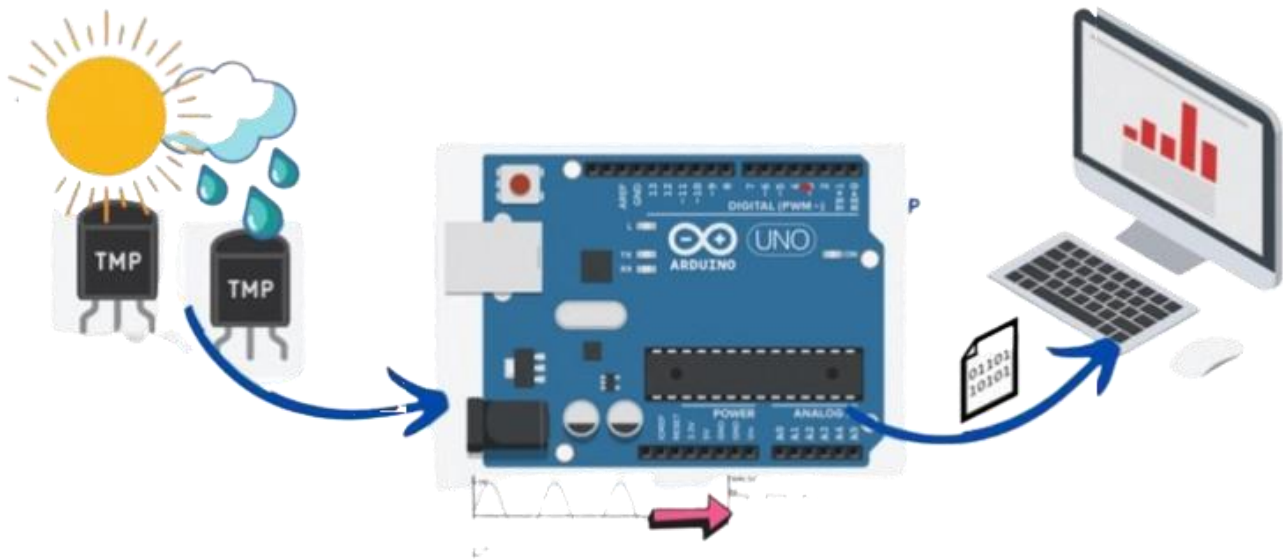


ARDUINO COURSE



**TEACHER:
CERTIFIED
TRAINER**

**DURATION:
16 CLASSES (1 HOUR PER CLASS)**

**MODE:
OFFLINE**

Arduino is an open-source electronics platform that enables students to build interactive and intelligent systems using simple coding and hardware. It introduces core concepts like microcontrollers, digital and analog signals, and circuit connections. Students learn to work with various sensors (ultrasonic, IR, LDR, temperature) and outputs like LEDs, buzzers, motors, and relays. Using the Arduino IDE, they write code to control these components and bring their ideas to life. The course is structured to build knowledge step by step—from basic circuits to complete automation projects. By the end of the program, students gain the confidence and skills to design and build their own custom robots.

Level 1: Foundations of Electronics & Motion (Ages 12–14)

1. Introduction to Arduino & LED Blink

- What is Arduino and how it works
- Upload code to blink an LED
- Simulate traffic lights using LEDs

2. Button-Controlled Buzzer

- Learn how input and output interact □
Press button to trigger buzzer alert

3. Temperature Sensor (DHT11) Monitoring

- Connect sensor and read data
- View temperature on Serial Monitor

4. Ultrasonic Sensor for Distance Sensing

- Measure distance using sensor
- Print distance values on screen

5. Servo Motor Control via Potentiometer

- Rotate servo based on knob position
- Prep for robotic walking movements

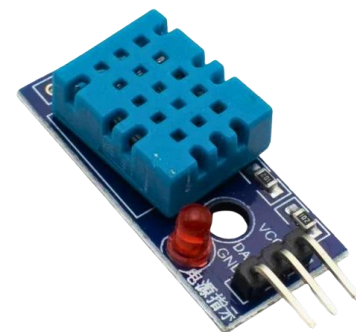
6. BO Motor & Driver Integration (L293D/L298N)

- Connect motor driver to Arduino
- Control DC motors in both directions

7. IR Sensor with Buzzer/LED Feedback

- Detect obstacle or black lines
- Trigger output when detection happens

Mega Projects (8 Sessions)



- **Hand Follower or Edge Follower Bot** – IR + motor control (2–3 sessions)
- **Obstacle Avoidance Car** – Ultrasonic + motor driver (2–3 sessions)
- **Hollow Clock with Stepper Motor** – Stepper motor and display logic (2 sessions)
- **Otto Bot (Walking Bot with Servo Motors)** – Assembly + servo logic (2–3 sessions)



Level 2: Intermediate Robotics & Smart Logic (Ages 14–16)

1. LDR + LED Indicators

- Detect light with LDR
- Automatically turn LEDs ON/OFF

2. Keypad Module

- A keypad module is used to enter a password.
- It helps access control access in security system.



3. IR Remote Control and receiver

- Use IR remote buttons
- Move servo or blink LEDs

4. Dual Motor Control with IR Remote

- Use remote to drive both motors □ Control forward, backward, turn

5. Flame Sensor with Alerts

- Sense fire or high heat
- Trigger buzzer and LED alert

6. Joystick-Controlled Servo

- Move servo in multiple directions
- Simulate robotic arm joints



7. OLED Display Integration

- Show values from sensors
- Display temperature or distance

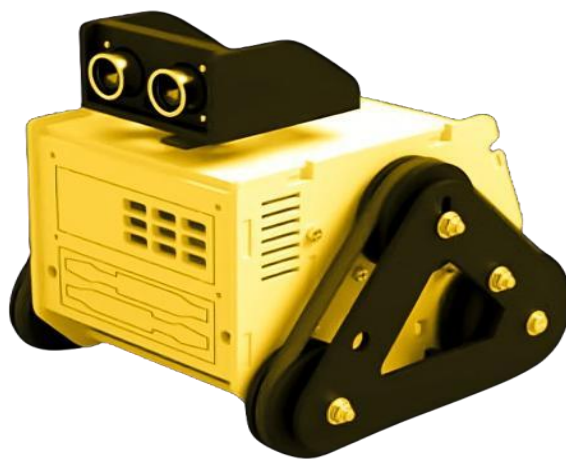
8. Sound Sensor (Clap Switch)

- Detect sound like a clap

- Toggle LED or motor on clap

Mega Projects

- **Wall-E (Tracking Robot)** – Line tracking + motor logic (2–3 sessions)
- **Fire-Fighting Robot** – Flame sensor + autonomous logic + fan (2–3 sessions)
- **Robotic Arm (Joystick or Pot Controlled)** – Multi-axis precision control (2–3 sessions)



Level 3: Advanced Automation & Wearables (Ages 16–18)

1. PIR Motion Detection Sensor

- Sense human movement
- Control light or alarm with motion

2. Relay Control for Appliances

- Operate fans, bulbs, or pumps □ Learn to switch AC/DC devices

3. Bluetooth-Based Control (HC-05)

- Connect Arduino to mobile
- Control robot or appliance remotely

4. Speaker with MIC and ISD1820PY Voice Recording Module

- It records and plays voice message.
- It is useful for alerts and security system.

5. Gas/Smoke Detection using MQ Sensors

- Detect gas or smoke leaks
- Create safety alert system

6. RFID tag

- A device that reads the data stored in the RFID tag using radio waves



Mega Projects

- **Home Automation System** – Use PIR, relay, Bluetooth, and display to create a smart home environment.
- **Bionic Hand (Flex Sensors & Servo Fingers)** – Wearable robotic hand with flex sensor input.

- **Dog or Cat Bot (Walking Pet Bot with Servos)** – Servo-controlled walking bot with personality, can include sound effects from the speaker module.

